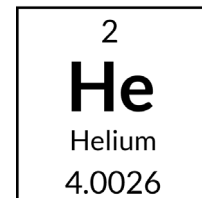
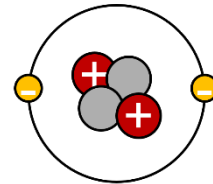


Key Learning Points: Atoms, Radiation, and Radioactivity

1. Atomic Structure

- **Atoms** are the smallest units of chemical elements, consisting of a nucleus (protons and neutrons) and electrons in defined shells.
- The **atomic number (Z)** defines the element and equals the number of protons (and electrons in a neutral atom).
- The **mass number (A)** is the sum of protons and neutrons.
- **Bohr's model** describes electrons in fixed energy shells; quantum numbers define shell size and electron allocation.
- **Electron binding energy** is the energy needed to remove an electron from its shell; it is highest for inner shells and increases with atomic number.



2. Energy and Mass

- **Einstein's equation** ($E = mc^2$) relates mass and energy, showing that a small amount of mass can be converted into a large amount of energy.
- **Electron volt (eV)** is the basic unit of energy in atomic physics.
- **Nuclear binding energy** is the energy required to split a nucleus into its components; it explains why some nuclei are more stable than others.

3. Radiation Types

Particulate Radiation

- Consists of particles (protons, neutrons, electrons, positrons) with mass and kinetic energy

Electromagnetic Radiation

- Includes x-rays and gamma rays, which are massless and travel at the speed of light
- Energy relates to wavelength and frequency

4. Radioactivity and Nuclear Stability

Nuclear Stability

- Depends on the balance between strong nuclear forces (attractive, between nucleons) and electromagnetic forces (repulsive, between protons)

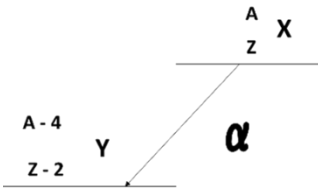
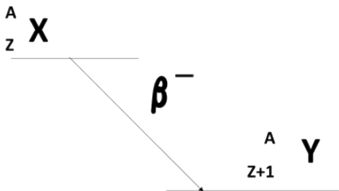
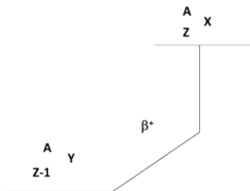
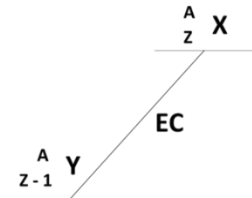
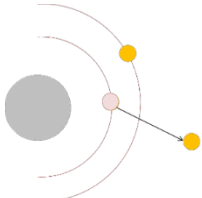
Stable Nuclei

- Have optimal neutron-to-proton ratios (1:1 for light elements; more neutrons for heavier elements)

Unstable Nuclei

- Undergo radioactive decay to achieve stability, emitting radiation in the process

5. Modes of Radioactive Decay

Alpha Decay (α)	Beta-minus Decay (β^-)	Beta-plus Decay (β^+ or Positron Emission)
Emission of a helium nucleus (2 protons, 2 neutrons); occurs in heavy nuclei.	A neutron converts to a proton, emitting an electron and an antineutrino.	A proton converts to a neutron, emitting a positron and a neutrino.
		Positron Decay Scheme 
Electron Capture	Gamma Decay	Internal Conversion
A proton converts to a neutron, emitting a positron and a neutrino.	The nucleus captures an inner electron, converting a proton to a neutron.	Excitation energy is transferred to an orbital electron, which is then ejected.
		

6. Characteristic and Auger Radiation

Characteristic X-rays

- Emitted when an electron from a higher shell fills a vacancy in a lower shell
- Releases energy equal to the difference in binding energies

Auger Electrons

- Emitted when the energy from an electron transition is transferred to another electron
- Ejects the other electron

7. Nuclear Nomenclature

Isotopes

- Same atomic number, different mass numbers
- Isotopes have the same number of protons

Isotones

- Same number of neutrons, different atomic numbers
- Isotones have the same number of neutrons

Isobars

- Same mass number, different atomic numbers
- Isobars have the same A number (mass number)

Isomers

- Same atomic and mass numbers, different energy states
- Isomers are the same molecules entirely rearranged

8. Applications in Medicine

- Many radioactive decay processes and isotopes are fundamental to diagnostic imaging (e.g., PET, SPECT) and radiation therapy.
- Understanding decay schemes and radiation types is crucial for safe and effective use in medical physics.